Assignment 03: MLflow Model Selection and Dockerized Inference Service

AI545 - W26

Ahsanul Kabir

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1 Training and Model Selection Process

The training and model selection phase was done using three Python scripts.

- $train.py$: Five distinct models (variations of Logistic Regression, Random Forest, and Gradient Boosting) were trained on the scikit-learn Breast Cancer dataset. Both accuracy and recall metrics were recorded for each run, alongside the model artifacts.
- $register_champion.py$: To programmatically select the best model, the *MlflowClient* was used to query the experiment runs, sorting them by $metrics.recall$ in descending order and fetching the top result. This top-performing model was then registered in the MLflow Model Registry under the name $BreastCancerChampion$ and assigned the *champion* alias.
- $export_model.py$: The champion model was loaded from the MLflow registry using its alias ($models:/BreastCancerChampion@champion$) and serialized into a standalone $model.pkl$.

2 Docker Containerization

This was done using $.dockerignore$ and $Dockefile$. $Dockefile$ was written as efficiently as possible. After building the image, the image was run for inference.

3 Commands

Terminal 1 for MLFlow

Terminal

```
mlflow ui
```

Terminal 2 for Python scripts and API calls

Terminal

```
python train.py
python register_champion.py
python export_model.py
curl -X POST http://localhost:8000/predict \
  -H  Content-Type: application/json  \
  -d '{
     features :  [17.99,10.38,122.8,1001.0,0.1184,0.2776,0.3001,0.1471,
                  0.2419,0.07871,1.095,0.9053,8.589,153.4,0.006399,0.04904,
                  0.05373,0.01587,0.03003,0.006193,25.38,17.33,184.6,2019.0,
                  0.1622,0.6656,0.7119,0.2654,0.4601,0.1189]
  }'
```

Terminal 3 for Docker commands and run

Terminal

```
docker build -t breast-cancer-inference .
docker run -p 8000:8000 --name inference-service breast-cancer-inference
```

4 Example API Request Response

```
curl -X POST http://localhost:8000/predict \
  -H "Content-Type: application/json" \
  -d '{
    "features": [17.99,10.38,122.8,1001.0,0.1184,0.2776,0.3001,0.1471,
                 0.2419,0.07871,1.095,0.9053,8.589,153.4,0.006399,0.04904,
                 0.05373,0.01587,0.03003,0.006193,25.38,17.33,184.6,2019.0,
                 0.1622,0.6656,0.7119,0.2654,0.4601,0.1189]
  }'
```

```
{
  "prediction": 0,
  "label": "malignant",
  "probabilities": {
    "malignant": 0.98,
    "benign": 0.02
  }
}
```

5 Screenshots

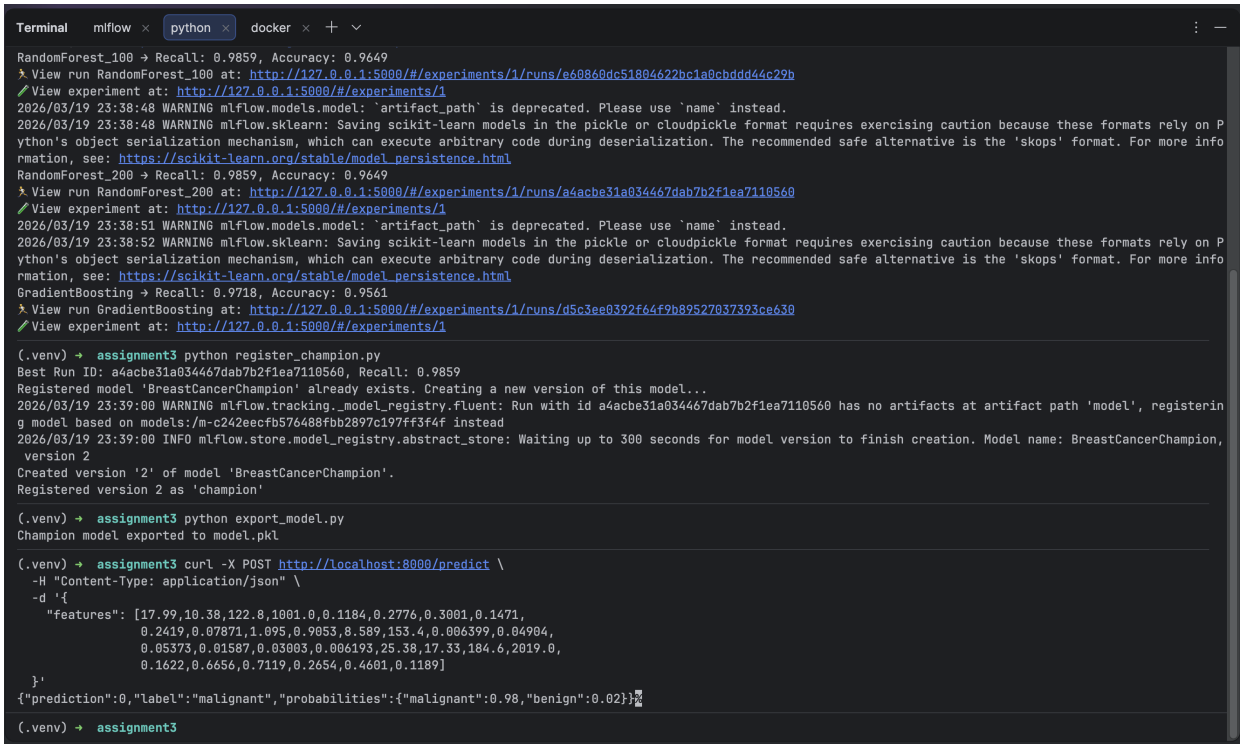

Figure 1: Python Scripts and cURL

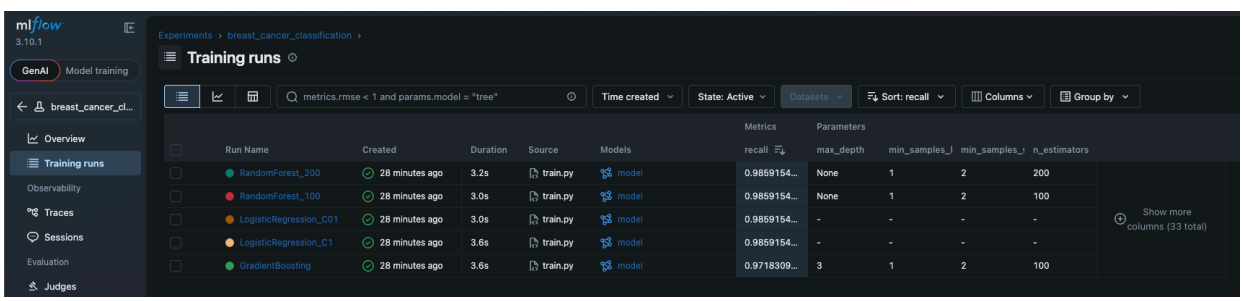

Figure 2: MLFlow Experiment Runs

```
Terminal    mlflow ×    python ×    docker ×

➜  assignment3 docker build -t breast-cancer-inference .
[+] Building 156.5s (11/11) FINISHED                                                                                    docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                                                    0.0s
 => => transferring dockerfile: 263B                                                                                                    0.0s
 => [internal] load metadata for docker.io/library/python:3.11-slim                                                                     1.1s
 => [internal] load .dockerignore                                                                                                       0.0s
 => => transferring context: 236B                                                                                                       0.0s
 => [1/6] FROM docker.io/library/python:3.11-slim@sha256:9358444059ed78e2975ada2c189f1c1a3144a5dab6f35bff8c981afb38946634              19.2s
 => => resolve docker.io/library/python:3.11-slim@sha256:9358444059ed78e2975ada2c189f1c1a3144a5dab6f35bff8c981afb38946634               0.0s
 => => sha256:d6c44f24ac6d3ed8e0f59c89c5dd485f80db5404d3710e5289d970d53889f49b 14.32MB / 14.32MB                                        7.3s
 => => sha256:ccdb4a613b5ce528e7385f571fafed3702a7390a0adbd685508a926f0db5f649 249B / 249B                                              0.2s
 => => sha256:b4822ed67dff6dee67342a10debf327642a6d8f6a65b7080bbf4d3d8e048e6fc 1.27MB / 1.27MB                                          2.5s
 => => sha256:f4badedbec24858ef2dc51256f6418328e305e9c3c5a5e093581f83425618bd5 30.14MB / 30.14MB                                        18.1s
 => => extracting sha256:f4badedbec24858ef2dc51256f6418328e305e9c3c5a5e093581f83425618bd5                                               0.7s
 => => extracting sha256:b4822ed67dff6dee67342a10debf327642a6d8f6a65b7080bbf4d3d8e048e6fc                                               0.1s
 => => extracting sha256:d6c44f24ac6d3ed8e0f59c89c5dd485f80db5404d3710e5289d970d53889f49b                                               0.3s
 => => extracting sha256:ccdb4a613b5ce528e7385f571fafed3702a7390a0adbd685508a926f0db5f649                                               0.0s
 => [internal] load build context                                                                                                       0.0s
 => => transferring context: 661.48kB                                                                                                   0.0s
 => [2/6] WORKDIR /app                                                                                                                  0.2s
 => [3/6] COPY requirements.txt .                                                                                                       0.0s
 => [4/6] RUN pip install --no-cache-dir -r requirements.txt                                                                          105.0s
 => [5/6] COPY app.py .                                                                                                                 0.2s
 => [6/6] COPY model.pkl .                                                                                                              0.0s
 => exporting to image                                                                                                                 30.7s
 => => exporting layers                                                                                                                26.4s
 => => exporting manifest sha256:70785ba537f9d6829dbe492e4df2863fe1ae69f45372a6d600953d58abd25dfb                                       0.0s
 => => exporting config sha256:f5413cb937296edc40a03986ef10d3a93ee048e848381dd775ad1f6fb3d33181                                         0.0s
 => => exporting attestation manifest sha256:b12a37e770530ae3b653628501ce62866d9826140bd397e2a9ed5b87fc6f448b                           0.0s
 => => exporting manifest list sha256:641333aa90877008c80224534afe80e4ed693d826b30bba86c3ff7b1c82f5d3f                                  0.0s
 => => naming to docker.io/library/breast-cancer-inference:latest                                                                       0.0s
 => => unpacking to docker.io/library/breast-cancer-inference:latest                                                                    4.3s

➜  assignment3 docker run -p 8000:8000 --name inference-service breast-cancer-inference

INFO:     Started server process [1]
INFO:     Waiting for application startup.
INFO:     Application startup complete.
INFO:     Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
INFO:     192.168.65.1:52958 - "POST /predict HTTP/1.1" 200 OK
```

Figure 3: Docker Commands